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WORK IN PROGRESS

GRAB : Griefing Recognition via Anomaly Behavior detection

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GRAB : Griefing Recognition via Anomaly Behavior detection

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Abstract

Trolling behavior in online games is a persistent issue that disrupts user experience or player enjoyment, and even leads to player churn. This paper proposes a deep learning-based method for detecting intentional toxic behavior, with a particular focus on griefing, by utilizing in-game telemetry data from PlayerUnknown's Battlegrounds (PUBG). Using behavioral patterns aggregated from a two-month data collection period, our model achieves strong anomaly detection performance while remaining interpretable for developers by anomaly score. We validate our approach through experimental analysis and expert feedback.

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1 Introduction

"Trolling" is a widespread issue in online games. Intentional trolling not only induces psychological harm, such as lower self-esteem in victims, but also increases the likelihood of retaliatory trolling, contributing to a cycle of toxic behavior.[22] [1] [20] However, trolling is not clearly defined and can be interpreted differently depending on the context, which makes it difficult to be detected through automated methods. [6] To address this issue, numerous game companies have developed and deployed automated detection systems designed to identify and manage trolling behavior in online environments. [17]

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Despite these efforts, existing detection approaches are often limited by subjectivity, susceptibility to abuse of the reporting system, and a high incidence of false positives. [2] To overcome these limitations, recent studies have explored automated detection methods based on in-game data. Most of these efforts focus on language-based features, such as chat logs or voice communication. [17] [23] Although effective in some contexts, such methods are less reliable in scenarios where player communication is limited or absent. [13]

In line with technological advances, artificial intelligence techniques have been increasingly incorporated into trolling detection systems to enhance accuracy and efficiency. [18] [11] However, AI-based systems often encounter challenges such as spurious correlations, scarcity of labeled data, and limited interpretability, which raise concerns regarding their reliability and robustness. [21].

This study focuses on a specific subtype of trolling known as griefing, which means intentionally harming teammates or sabotaging the game by in-game action.[14] It is often difficult to detect, especially in nonverbal gameplay contexts, so we propose an anomaly detection framework based on deep learning designed to identify griefing behaviors based on in-game behavioral data.

To achieve this goal, DevNet model, which can be trained with only a small amount of labeled data(semi-supervised learning) for anomaly detection, is utilized. [19] Unlike prior research that primarily relied on linguistic cues or heuristic methods, our study analyzes in-game behavior by deep learning model, using aggressive and cooperative indicators in light of game design.

2 Related Work

Assessing the impact of griefing behaviors on players' psychological needs, Achterbosch et al. [1] found that increased griefing incidents consistently reduced victims' autonomy and relatedness, highlighting the need for effective victim protection mechanisms.

Cook et al. [8] conducted interviews with 22 identified trolls, categorizing their behaviors into verbal and behavioral types, including team disruption, contrary play, and aiding enemies. This study also revealed the cyclical nature of trolling, noting victims or bystanders could become future trolls themselves.

Research on verbal toxicity detection, such as message and voice chat trolling, has been actively pursued. For example, Morrier et al. [18] used a reinforcement learning based on proximal policy optimization (PPO) to detect toxic message or chat, achieving superior

performance over traditional supervised and rule-based approaches due to its adaptive capabilities and responsiveness.

However, two key issues persist: (1) privacy - raw voice can expose biometric or personal data, and (2) fragility of speech recognition - noisy multispeaker chats hurt transcription accuracy and slow moderation, as noted by Jiang et al. [12]. Thus, even though chat-based detection has been improved using SOTA AI models, it remains limited by language and context [25].

Studies have also explored behavioral indicators in game in terms of griefing. Canossa et al. [3] demonstrated that gameplay data alone could effectively identify toxic players. Using 36 in-game features, their Random Forest model achieved 88% accuracy. However, the study utilized labeled datasets provided by Ubisoft involving heuristic labeling methods, limiting its generalizability despite demonstrating strong potential for broader genre application.

In another gameplay-focused study, Blackburn and Kwak [2] automated League of Legends' Tribunal system—previously dependent on human judgments—using a Random Forest model. Despite significantly reduced operational time and costs, the Tribunal system was eventually discontinued due to inherent subjectivity.

Chen et al. [5] proposed a visual analytics system designed to efficiently identify grievers in MOBA games, reducing expert identification time from approximately 25 minutes to 5 minutes. However, final decisions still required human judgment, presenting a notable limitation.

3 Methodology

In online games, griefers behave differently from regular players by intentionally hindering gameplay rather than contributing to the team's success. This disruptive behavior results in distinct gameplay patterns that differentiate them from normal users. In this study, we propose a detection framework that identifies griefing behavior by analyzing behavioral differences between griefers and regular players, as informed by pre-interviews. The proposed framework consists of four main stages:

- (1) **Game Log Collection:** Gameplay log data is collected using the official PUBG Open API, which provides time-stamped, match-level records of player actions. The collected data is stored for further analysis.
- (2) **Data Preprocessing:** Raw log data is transformed into structured vectorized representations suitable for training deep learning models. This involves aggregating player behavior events into numerical features that summarize cooperation and disruption indicators.
- (3) **Behavioral Analysis and Semi Labeling:** Based on interviews with experienced FPS game developer and active PUBG player, we identify key behavioral traits that differentiate griefer from regular players. Features are selected according to these insights, and a labeling scheme is implemented using a heuristic based on the ratio of disruptive to cooperative actions (e.g., team attacks, vehicle co-rides).
- (4) **Model Training and Evaluation:** A subset of the dataset is sampled and split into training and testing sets. An anomaly detection model is trained on the weakly labeled data to learn the patterns associated with griefing behavior, and its performance is validated using test dataset.

In this study, we employed an anomaly detection model [4]. An anomaly detection model is designed to identify data instances that deviate significantly from the normal pattern. When a new input x is provided, the model (DevNet) computes an anomaly score $f(x)$ that quantifies how much the instance differs from typical behavior. The greater the score, the more likely the instance is to be anomalous.

A predefined threshold θ is used to make the decision: if $f(x) > \theta$, the instance is classified as anomalous; otherwise, it is considered normal (Fig 1). This approach enables the model to operate without explicit class labels, relying instead on learned deviations from known distributions of normal behavior. Anomaly detection is particularly useful in scenarios where labeled abnormal samples are scarce or unavailable. Among various anomaly detection models, we selected **DevNet**, which has been shown to perform well even with limited labeled data.[19][16]



Figure 1: An illustration of anomaly detection process using a threshold-based decision rule.

4 Experiment

4.1 Data Collection

Gameplay log data were collected through the official API provided by Krafton, the developer of the battle royale game PlayerUnknown's Battlegrounds (PUBG). This study focused exclusively on match data from the Steam platform, covering the period from April 7 to May 7, 2025. A total of 26,807 matches were retrieved during this time frame. Each log captures time-stamped actions within a match, recording the in-game behavior of individual players.

4.2 Data Preprocessing

To prepare the collected log data for training with the DevNet anomaly detection model, data preprocessing was necessary. Behavioral counts were aggregated for each player, representing how frequently specific in-game actions occurred. Given the study's focus on griefing behavior, particularly team-killing, we filtered the dataset to include only teams where at least one teamkill event occurred. As a result, a total of 28,000 player data were collected, where each row represented one player in a single match.

4.3 Interviews

The griefing behaviors differ significantly among individuals. Moreover, the PUBG log data used in this research do not contain ground truth labels such as prior reports or confirmed cases of trolling. To construct an appropriate feature set and apply a pseudo-labeling approach, we conducted exploratory interviews with a former PUBG developer (now an FPS game developer) and a PUBG player with over 800 hours of playtime and 7 years of experience.

To systematically identify representative griefing behaviors in PUBG, we extracted and categorized recurring themes mentioned

by both interviewees. While responses varied in form, several actions were consistently described as disruptive, intentionally harmful, or misaligned with the game’s survival objectives. These included behaviors such as deliberate team damage, obstructive item use, and intentional disengagement from team strategy. From these repeated mentions, we compiled a categorized list of behaviors, which is summarized in Table 1.

Based on the interviews, the following behaviors were identified as representative forms of griefing in PUBG (Table 1). These behaviors were repeatedly emphasized by the interviewees as disruptive to cooperative gameplay and counter to the intended survival objectives of the game. For example, deliberately damaging or killing teammates using firearms or vehicles (entries 2 and 3) was consistently described as a highly recognizable and quantifiable form of griefing that often leaves strong traceable evidence in server logs. As such, we prioritized these behaviors as the primary focus of this study’s modeling approach.

Other behaviors listed in Table 1, such as AFK behavior, solo landing, or misuse of throwables, were also mentioned in the interviews. However, these actions were considered to be more ambiguous in terms of intentionality, or difficult to isolate reliably from log data alone without additional context such as chat logs or player location clustering. Therefore, while such behaviors helped shape the theoretical understanding of griefing, they were not selected as core detection targets due to limitations in objective measurability and reproducibility.

No.	Trolling Behavior Description
1	Landing at a location different from the rest of the team and playing solo
2	Inflicting damage or killing teammates using guns or throwable weapons
3	Intentionally hitting teammates with a vehicle to cause damage
4	Using throwable items such as flashbangs or smoke grenades to obstruct teammates
5	Deliberately committing suicide or intentionally dying to the enemy
6	AFK (Away From Keyboard) behavior
7	Using abusive language through voice or text chat

Table 1: Summary of trolling behaviors in PUBG based on interviews

Among these scenarios, team-killing was identified as the most explicit and quantifiable form. Nevertheless, it was also acknowledged that some team damage may result from accidental or low-skill play, requiring careful consideration of the surrounding context.

To distinguish between intentional and unintentional griefing, interviewees recommended examining both the method of attack and its contextual circumstances—such as vehicle-sharing patterns or distance from teammates. These recommendations led us to incorporate cooperative gameplay indicators alongside attack-type features when constructing variables and pseudo-labels for model training and evaluation. However, both interviewees agreed that

the number of reviving teammates cannot be considered as explicit cooperative or trolling behavior considering the open-world trait of PUBG.

4.4 Feature Selection

Through expert interviews, we confirmed that distinguishing between cooperative and non-cooperative behavioral indicators is effective in differentiating griefer from regular players. Especially, the number of attacking teammates is thought to be most non-cooperative and maintaining close distance with teammates is thought to be quantitative and explicit cooperative behavior, which is represented by vehicle co-riding counts.

Therefore, to enable quantitative analysis, we selected relevant features representing cooperative and non-cooperative behavior. The number of team attacks, categorized by weapon type, was used as an indicator of non-cooperative behavior, while the number of teammate revives and vehicle co-rides were chosen as indicators of cooperative behavior. This decision was grounded in prior findings from Sauvik Das et al[9], who demonstrated that players with higher revive counts tend to be more cooperative and engaged, and also die less frequently during gameplay. The frequency of vehicle co-riding with teammates was regarded as an explicit indicator of cooperative behavior, as it reflects spatial alignment and coordination with team members. Based on the aforementioned interview insights, the revive count was excluded during the semi-labeling process and used solely for model training.

To ensure the model remained efficient and scalable, we excluded irrelevant or redundant variables that could increase data dimensionality. Only features that clearly reflected griefing characteristics—based on both interview insights and empirical evidence—were retained. These selected variables were then used to train the detection model, (as described in Appendix B)

4.5 Data Semi Labeling

Based on insights from prior interviews, we hypothesized that griefers would exhibit more non-cooperative than cooperative behaviors. To operationalize this, we used the number of team attacks as a proxy for non-cooperative actions and the number of co-rides in vehicles with teammates as an indicator of cooperative behavior.

We then defined a labeling metric as the ratio of team attacks to vehicle co-rides. Since the dataset was constructed from matches involving at least one instance of team-killing, we provisionally labeled the top 25% of users with the highest ratio values as griefers(PUBG’s squad game has 4 people in a team). This initial labeling was later revised based on the model’s performance during validation.

4.6 DevNet Experiment

This study employed the DevNet model, which can detect anomalous users in the entire dataset by utilizing a small subset of labeled data(Semi-labeled data)Pang et al. [19]. Unlike traditional machine learning-based anomaly detection methods, DevNet is an end-to-end neural network model that computes anomaly scores directly from the input data. DevNet is trained using a deviation loss based on Z-score, ensuring that labeled anomalous samples have statistically significantly higher anomaly scores compared to normal

samples, relative to the average anomaly score of labeled anomalies [19]. Consequently, DevNet allows for statistically meaningful anomaly score calculations; however, since DevNet does not directly assign labels, determining an appropriate threshold is essential.

The DevNet neural network structure used in this study consists of multilayer perceptron (MLP) layers, with 3 hidden layers, Batch Normalization, ReLU activation functions, and Dropout (rate of 0.3) applied between each layer. The architecture of the model implemented in this study closely follows the structure proposed in the original DevNet paper. During training, a deviation loss function utilizing binary labels (anomaly: 1, normal: -1) was employed. The model was trained for 200 epochs with an Adam optimizer and a learning rate of 0.001. To avoid linear dependency issues, the input features used for temporary labeling (vehicle ride-sharing frequency and team attack frequency) were excluded from the model inputs.

The experimental dataset comprised a total of 28,000 instances, with approximately 7,000 instances (top 25%) temporarily labeled as intentional troll users based on preliminary interview insights. This criterion was set at 25% because data collection focused exclusively on teams with at least one team-killing player. From the approximately 7,000 temporarily labeled instances, 50% (about 3,500 instances) were separated as the test set for model evaluation, while the remaining labeled 3,500 instances and approximately 21,000 unlabeled normal user instances formed the training set, totaling around 24,500 instances.

As shown in Figure 2, the DevNet loss at 200 epochs was significantly lower than at initialization. After training DevNet on the training set, anomaly scores (DevNet Scores) were computed for each data point within the training set. As previously mentioned, since DevNet Scores alone do not provide explicit labels, an appropriate percentile threshold had to be established to distinguish between troll and normal users. Subsequently, the same DevNet model was applied to the separate test set (approximately 3,500 instances) to calculate anomaly scores. Accuracy was determined by applying the established percentile threshold from the training set to assess how effectively the model identified intentional troll users labeled according to preliminary research findings.

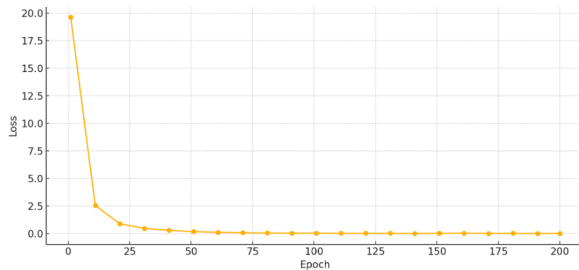


Figure 2: DevNet Loss at 200 epochs

5 Results

The experimental results from the training set, analyzing DevNet Scores at percentile thresholds ranging from 50 to 90 in increments of 0.5, are as follows in figure 3. The accompanying graph demonstrates a gradual increase in DevNet Scores between approximately

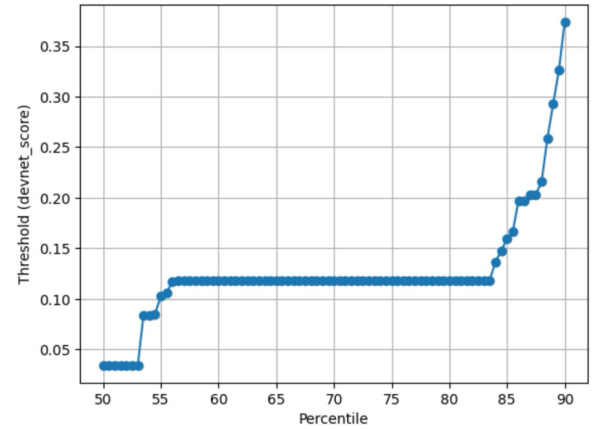


Figure 3: Threshold at each percentile from 50 to 90

the 55th and 85th percentiles, followed by a sharp increase at the 87.5th percentile. Users within this percentile range exhibited similar levels of DevNet Scores. Thus, users scoring above approximately 0.203156 at the 87.5th percentile can be suspected as intentional trolls. Among the data points exceeding this threshold, approximately 91.4% were also classified as intentional trolls during the semi-labeling process.

Therefore, the training set successfully enabled modeling of DevNet Scores and identification of an appropriate threshold for anomaly detection. To validate the derived model and threshold, the separate test set (consisting entirely of temporarily labeled intentional troll users) was analyzed. After applying the DevNet model to this test set, approximately 71.75% of data points exceeded the threshold DevNet Score of 0.203156 identified from the training set.

6 Discussion

The results of this study demonstrate the potential of semi-supervised deep learning approach for detecting intentional disruptive behavior – specifically griefing – in an online team-based game known as PUBG. However, several important considerations emerge when interpreting these findings and thinking about their practical application.

Although the DevNet model shows fairly strong performance using minimal label, the interpretability and trustworthiness of its anomaly scores warrant further investigation. In particular, it is important to ensure that high anomaly scores truly reflect malicious intent than outlier behavior due to unconventional playstyles or low skill. Currently, our model is highly sensitive to cooperative and disruptive behaviors such as team damage and revive avoidance. However, this narrow focus may overlook other trolling tactics that don't involve direct attacks as mentioned by experts in preliminary interviews. The practical application of our findings should also be discussed. Detecting troll behavior is only the first step; what actions should follow such a detection is a more delicate issue. Companies may consider account bans or matchmaking restrictions, but these punitive measures might be disproportionate responses, especially when the underlying behavior could be

accidental or situational [7]. A more nuanced approach could involve 'troll quarantine zones', which is a segregated matchmaking pool where players flagged for repeated toxic behavior are grouped together. This method protects regular players from being paired with troll players while minimizing the risk of unfairly penalizing innocent players. Additionally, it is crucial to ensure that such systems do not penalize low-skilled or atypical players unfairly, which requires continuous refinement of both detection methods and response strategies.

7 Limitations and Future Work

The current framework detects griefing behavior based on aggregated log data, focusing primarily on the statistical outcomes of gameplay actions such as team-kills. However, preliminary interviews with experienced developers and players revealed that griefing behaviors manifest in a variety of forms, which are often highly dependent on the in-game context and temporal flow of actions. Examples include going AFK, committing suicide intentionally, obstructing teammates using flashbangs or smoke grenades, and deliberately separating from the team during landing to act independently. These behaviors are not easily captured by simple frequency-based metrics, as they require an understanding of the behavioral sequence and its surrounding context.

To address this limitation, future work will focus on modeling player behavior as time-ordered sequences rather than static aggregates. We plan to convert raw gameplay logs into sequential vector representations that preserve the temporal structure of player actions. These sequences will then be analyzed using deep learning models designed for temporal data, such as Long Short-Term Memory (LSTM) networks or Transformer-based architectures [24] [15]. By incorporating temporal dynamics and contextual transitions, we aim to improve the model's ability to detect complex and subtle griefing patterns, ultimately enhancing both accuracy and interpretability. Lastly, the findings of this experiment are limited to PUBG, which is an FPS game, reducing its applicability to other types of online game such as MMORPGs, MOBAs, etc. Different genre games exhibit different interaction patterns and player norms, which may affect the nature and expression of grief behaviors of troll players [10] [13]. Future research should be conducted on proposing a framework for multiple game environments, adapting features and model parameters accordingly.

8 Conclusion

We propose a semi-supervised framework to detect intentional toxic behavior in an online game, PUBG, using structured in-game logs. This framework can be considered effective for capturing the intentionality of griefing behavior, as it is trained on semi-labeled data derived from the most explicit forms of trolling identified through qualitative analysis. Our method reduces the reliance on labeled data and captures behavioral signals aligned with known grief patterns. This contributes to scalable and interpretable troll detection systems for developers.

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Appendix

A Implementation Code

The implementation code is available at: https://github.com/Jewoo-Park/GRAB_Griefing_Recognition_via_Anomaly_Behavior_detection/tree/main

B Behavioral and weapon-based features selected for training the detection model

Feature	Description
victim count	Number of times a player was revived by teammates in a match
revive count	Number of times a player revived teammates in a match
vehicle count	Number of times a player shared a vehicle with teammates in a match
Damage Gun	Number of times a player attacked a teammate using a firearm
Damage Explosion Grenade	Number of times a player attacked a teammate using a grenade
Damage VehicleHit	Number of times a player inflicted damage to a teammate using a vehicle
Damage Melee	Number of times a player attacked a teammate using melee attacks
Damage MeleeThrow	Number of times a player attacked a teammate using thrown melee weapons
Damage Explosion JerryCan	Number of times a player attacked a teammate using JerryCan explosions
Damage Molotov	Number of times a player attacked a teammate using molotovs
Damage BlueZone-Grenade	Number of times a player attacked a teammate using BlueZone grenades
Damage Punch	Number of times a player attacked a teammate using punches
Damage Explosion PanzerFaustWar-head	Number of times a player attacked a teammate using PanzerFaust warhead explosion
Damage Explosion C4	Number of times a player attacked a teammate using C4 explosion
Damage Explosion Vehicle	Number of times a player attacked a teammate using vehicle explosion
Damage Explosion Mortar	Number of times a player attacked a teammate using mortar explosion
Damage Explosion PanzerFaustBack-Blast	Number of times a player attacked a teammate using PanzerFaust backblast
Damage Explosion StickyBomb	Number of times a player attacked a teammate using sticky bomb explosion
Damage DropStruc-tHit	Number of times a player attacked a teammate using falling structures

Table 2: Behavioral and weapon-based features selected for training the detection model